Subject: Mathematics | **WEEKLY CHAPTER TEST – 01 (PROBABILITY)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks: 20**

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

Which of the following cannot be the probability of an event?

(a) $\frac{2}{3}$ (b) -1.5

(c) 15% (d) 0.7

Working & Explanation:

- By the fundamental axiom of probability, the probability $P(E)$ of any event E must satisfy:

$$0 \leq P(E) \leq 1$$

- Option (b) is $-1.5 < 0$, which is negative and therefore **cannot** represent a probability.
- Options (a) $\frac{2}{3} \approx 0.67$, (c) $15\% = 0.15$, and (d) 0.7 all lie strictly in $[0, 1]$ and are valid probabilities.

CORRECT ANSWER: OPTION (b) $\longrightarrow$ -1.5

Question 2 [1 Mark]

Two unbiased coins are tossed simultaneously. The probability of getting at most one head is:

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{3}{4}$ (d) 1

Working & Explanation:

- Complete sample space of two simultaneous coin tosses:

$$S = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\} \implies n(S) = 4$$

- The event "at most one head" includes outcomes with 0 heads or 1 head:

$$E = \{\text{HT}, \text{TH}, \text{TT}\} \implies n(E) = 3$$

- Therefore, by classical probability formula:

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}$$

CORRECT ANSWER: OPTION (c) $\longrightarrow$ $\frac{3}{4}$

Question 3 [1 Mark]

A card is drawn from a well-shuffled deck of 52 cards. The probability of getting a red face card is:

(a) $\frac{3}{13}$ (b) $\frac{3}{26}$

(c) $\frac{1}{26}$ (d) $\frac{3}{52}$

Working & Calculation:

- Total cards in a standard deck: $n(S) = 52$.
- Red face cards (Kings, Queens, Jacks of Hearts and Diamonds):

$$n(E) = 2 \text{ suits} \times 3 \text{ face cards} = 6 \implies P(\text{red face card}) = \frac{n(E)}{n(S)} = \frac{6}{52} = \frac{3}{26}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow \frac{3}{26}$

Question 4 [1 Mark]

A bag contains 3 red balls, 5 black balls, and 4 white balls. A ball is drawn at random from the bag. The probability of not getting a black ball is:

(a) $\frac{5}{12}$ (b) $\frac{7}{12}$

(c) $\frac{1}{3}$ (d) $\frac{1}{4}$

Working & Calculation:

- Total balls in the bag: $n(S) = \text{Red} + \text{Black} + \text{White} = 3 + 5 + 4 = 12$.
- Favourable outcomes (drawing either a red or white ball):

$$n(\text{not black}) = 3 + 4 = 7 \implies P(\text{not a black ball}) = \frac{n(\text{not black})}{n(S)} = \frac{7}{12}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow \frac{7}{12}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): The probability of an impossible event is 0.
Reason (R): The probability $P(E)$ of any event E always lies between 0 and 1, i.e., $0 \leq P(E) \leq 1$.

Options (Each on an individual line for complete readability):

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

Step-by-Step Logical Analysis:

1. **Assertion (A):** An impossible event has $n(E) = 0 \implies P(E) = \frac{0}{n(S)} = 0. \implies$ **(A) is TRUE.**
2. **Reason (R):** By axiomatic definition, $0 \leq P(E) \leq 1$ universally. $\implies$ **(R) is TRUE.**
3. **Explanation Check:** (R) states the range of probabilities, but does not explain *why* the impossible event is 0 (which is because $E = \emptyset$). Hence, (R) is **not** the correct explanation of (A).

CORRECT ANSWER: OPTION (b) $\longrightarrow$ Both (A) and (R) are true, but (R) is not the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

Question 6 [2 Marks]

Cards each marked with one of the numbers 4, 5, 6, ..., 20 are placed in a box and mixed thoroughly. One card is drawn at random. What is the probability of getting an even number?

Step-by-Step Solution:

- **Step 1: Total number of possible outcomes, $n(S)$**
  The numbers marked on cards are $4, 5, 6, \ldots, 20$:

$$n(S) = \text{Last number} - \text{First number} + 1 = 20 - 4 + 1 = 17 \text{ cards}$$

- **Step 2: Favourable outcomes (Even numbers), $n(E)$**
  Let E be the event of drawing an even number: $E = \{4, 6, 8, 10, 12, 14, 16, 18, 20\}$.
  In arithmetic progression ($a = 4, d = 2, l = 20$):

$$n(E) = \frac{20 - 4}{2} + 1 = 8 + 1 = 9 \text{ outcomes}$$

- **Step 3: Calculate the probability**

$$P(E) = \frac{\text{Number of favourable outcomes } n(E)}{\text{Total number of possible outcomes } n(S)} = \frac{9}{17}$$

FINAL ANSWER: $\mathbf{P(\text{even number}) = \frac{9}{17}}$ [Total cards: 17, Even cards: 9]

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

Question 7 [3 Marks]

Two unbiased dice are thrown simultaneously. Find the probability of getting:

Sample Space: Total outcomes for two unbiased dice:

$$n(S) = 6 \times 6 = 36 \text{ ordered pairs } (x, y) \text{ where } x, y \in \{1, 2, 3, 4, 5, 6\}$$

(i) A sum of 8 [1 Mark]

Let E_1 be the event of getting a sum of 8 ($x + y = 8$). Favourable pairs:

$$E_1 = \{(2,6), (3,5), (4,4), (5,3), (6,2)\} \implies n(E_1) = 5 \implies P(E_1) = \frac{n(E_1)}{n(S)} = \frac{5}{36}$$

Answer (i): $\mathbf{P(\text{sum of } 8) = \frac{5}{36}}$

(ii) A sum greater than 9 [1 Mark]

Let E_2 be the event of getting a sum strictly greater than 9 (sum $\in \{10, 11, 12\}$):
Sum = 10: $\{(4,6), (5,5), (6,4)\}$ [3] | Sum = 11: $\{(5,6), (6,5)\}$ [2] | Sum = 12: $\{(6,6)\}$ [1]

$$n(E_2) = 3 + 2 + 1 = 6 \implies P(E_2) = \frac{n(E_2)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

Answer (ii): $\mathbf{P(\text{sum } > 9) = \frac{1}{6}}$

(iii) An even sum [1 Mark]

The sum $x + y$ is even when both dice show odd ($3 \times 3 = 9$) or both show even ($3 \times 3 = 9$):

$$n(E_3) = 9 + 9 = 18 \implies P(E_3) = \frac{n(E_3)}{n(S)} = \frac{18}{36} = \frac{1}{2}$$

Answer (iii): $\mathbf{P(\text{even sum}) = \frac{1}{2}}$

Question 8 [3 Marks]

All the red face cards are removed from a pack of 52 cards. The remaining cards are well shuffled and a card is drawn at random. Find the probability of getting:

Analysis of Remaining Cards:

- Initial deck = 52 cards. Red face cards removed: 2 suits × 3 face cards (K, Q, J) = 6 cards.
- Total remaining cards in deck: $n(S) = 52 - 6 = 46$ cards.

(i) A red card [1 Mark]

Original red cards = 26. Removed red face cards = 6 $\implies$ Remaining red cards = $26 - 6 = 20$.

$$P(\text{red card}) = \frac{\text{Number of remaining red cards}}{n(S)} = \frac{20}{46} = \frac{10}{23}$$

Answer (i): P(red card) = $\frac{10}{23}$

(ii) A queen [1 Mark]

Original queens = 4. Red queens removed = 2 $\implies$ Remaining queens = $4 - 2 = 2$ (Queen of Spades, Clubs).

$$P(\text{a queen}) = \frac{\text{Number of remaining queens}}{n(S)} = \frac{2}{46} = \frac{1}{23}$$

Answer (ii): P(a queen) = $\frac{1}{23}$

(iii) An ace of hearts [1 Mark]

Aces are number cards, not face cards. The Ace of Hearts was not removed and remains in the deck (1 card).

$$P(\text{an ace of hearts}) = \frac{1}{n(S)} = \frac{1}{46}$$

Answer (iii): P(an ace of hearts) = $\frac{1}{46}$

Question 9 [3 Marks]

A jar contains 24 marbles, some are green and others are blue. If a marble is drawn at random from the jar, the probability that it is green is $\frac{2}{3}$.

Given Data & Setup: Total marbles $n(S) = 24$. Let green marbles = g and blue marbles = b $\implies$ $g + b = 24$. Given $P(\text{green}) = \frac{2}{3}$.

(i) Find the number of green marbles in the jar [1 Mark]

$$P(\text{green}) = \frac{g}{n(S)} \implies \frac{g}{24} = \frac{2}{3} \implies g = 24 \times \frac{2}{3} = 16 \text{ green marbles}$$

Answer (i): Number of green marbles = 16

(ii) Find the number of blue marbles in the jar [1 Mark]

$$b = 24 - g = 24 - 16 = 8 \text{ blue marbles}$$

Answer (ii): Number of blue marbles = 8

(iii) What is the probability of drawing a blue marble? [1 Mark]

$$P(\text{blue}) = \frac{b}{n(S)} = \frac{8}{24} = \frac{1}{3} \quad \text{or via complement:} \quad P(\text{blue}) = 1 - P(\text{green}) = 1 - \frac{2}{3} = \frac{1}{3}$$

Answer (iii): P(blue marble) = $\frac{1}{3}$

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis [4 Marks]

Case Study Scenario:
A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 marked on a circular wheel, as shown in the carnival stall. Each of the 8 outcomes is equally likely. Avni decides to play the game by paying an entry token.
Sample Space: $S = \{1, 2, 3, 4, 5, 6, 7, 8\} \implies n(S) = 8$.



Circular Wheel Spinner · Arrow spins freely with 8 equally likely outcomes

(i) What is the probability that the arrow points at an odd number? [1 Mark]
The odd numbers in S are $E_1 = \{1, 3, 5, 7\} \implies n(E_1) = 4$.

$$P(\text{odd number}) = \frac{n(E_1)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Answer (i): $\mathbf{P(\text{odd number}) = \frac{1}{2}}$

(ii) What is the probability that the arrow points at a prime number? [1 Mark]
Prime numbers in $S = \{2, 3, 5, 7\} \implies n(E_2) = 4$ (Note: 1 is neither prime nor composite).

$$P(\text{prime number}) = \frac{n(E_2)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Answer (ii): $\mathbf{P(\text{prime number}) = \frac{1}{2}}$

(iii) What is the probability that the arrow points at a number greater than 2? [1 Mark]
Numbers strictly greater than 2 in $S = \{3, 4, 5, 6, 7, 8\} \implies n(E_3) = 6$.

$$P(\text{number } > 2) = \frac{n(E_3)}{n(S)} = \frac{6}{8} = \frac{3}{4}$$

Answer (iii): $\mathbf{P(\text{number } > 2) = \frac{3}{4}}$

(iv) What is the probability that the arrow points at a number less than 9? [1 Mark]
Every number $\{1, 2, 3, 4, 5, 6, 7, 8\} < 9$. This is a **sure (certain) event**:

$$E_4 = \{1, 2, 3, 4, 5, 6, 7, 8\} = S \implies n(E_4) = 8 \implies P(\text{number } < 9) = \frac{8}{8} = 1$$

Answer (iv): $\mathbf{P(\text{number } < 9) = 1}$

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